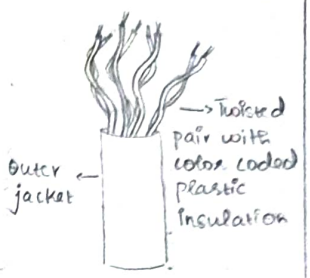
Ex. 02
6.8.26

Types of network cable.

AIM: Study of different types of network cables

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use	Image
UTP	Category 3 Category 5 Category 5e	10 bps upto 100 Mbps 1 Gbps	Adv: • Cheaper in cost • Easy to install as they have a smaller overall diameter Disadv: • More prone to (EMI) Electromagnetic interference & noise	10 Base-T Ethernet Fast Ethernet Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
STP SSTP	Category 6, 6a Category 7	10 Gbps 10 Gbps	Adv: • Shielded • Faster than UTP • Less susceptible to noise and interference Disadv: • Expensive • Greater installation effort	Gigabit Ethernet 10G Ethernet (55m) Widely used in data centres Gigabit Ethernet, 10G Ethernet (100m)	 
Coaxial cable	RG-6 RG-59 RG-11	10 - 100 Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile Disadv: • Limited distance • Cost • Size is bulky.	Speed of signal is 500 m Television network High speed internet connections.	
fibre optics cable	Single mode Multiple mode	100 Gbps	Adv: • High speed • High bandwidth • High security • Long distance Disadvantages: • Expensive • Requires skilled installers.	• Maximum distance of fibre optics cable is around 100 metres	

RESULT:
Thus the study on different types of network cables was done.